
1. Binary Clock

Program Name: Binary.java

Input File: binary.dat

Judges Input File

6

```
* 0 *  
00**0  
0*000  
000***  
0 0 0  
0*0**  
*0*0*  
*0**0*  
0 * *  
*0000  
00000  
00000*  
0 0 0  
00000  
00000  
00000*  
0 * 0  
0*00*  
000*0  
*****0  
* 0 0  
00000  
0*0*0  
0***00
```

Judges Output to Screen

```
08:25:59  
12:53:47  
04:08:09  
00:00:01  
11:59:34  
09:31:20
```

2. Cell Phone

Program Name: Cell.java Input File: cell.dat

Judges Input File

14.5
12.0
13.2
3.56
4.12
0.8
1.1
2.0
9.29
2.0
1.1
12.3
23.4
56.2
0.2
1.3
2.0
23.5
67.59
2.36

Judges Output to Screen

PLAN A: 247
PLAN B: 263

3. Change You Can Believe In

Program Name: Change.java

Input File: change.dat

Judges Input File

10
62
48
17
87
7
12
25
99
1
11

Judges Output to Screen

77
39
6
187
2
4
13
252
1
4

4. Elite Numbers

Program Name: Elite.java Input File: elite.dat

Judges Input File

10
33
85
1053
500
727
1499
771
901
601
1345

Judges Output to Screen

ELITE
NOT ELITE
ELITE
NOT ELITE
ELITE
ELITE
NOT ELITE
NOT ELITE
ELITE
ELITE

For Judges Information - Elite Numbers

[1, 3, 7, 9, 13, 15, 21, 25, 31, 33, 37, 43, 49, 51, 63, 67, 69, 73, 75, 79, 87, 93, 99, 105, 111, 115, 127, 129, 133, 135, 141, 151, 159, 163, 169, 171, 189, 193, 195, 201, 205, 211, 219, 223, 231, 235, 237, 241, 259, 261, 267, 273, 283, 285, 289, 297, 303, 307, 319, 321, 327, 331, 339, 349, 357, 361, 367, 385, 391, 393, 399, 409, 415, 421, 427, 429, 433, 451, 463, 475, 477, 483, 487, 489, 495, 511, 517, 519, 529, 535, 537, 541, 553, 559, 577, 579, 583, 591, 601, 613, 615, 619, 621, 631, 639, 643, 645, 651, 655, 673, 679, 685, 693, 699, 717, 723, 727, 729, 735, 739, 741, 745, 769, 777, 781, 787, 801, 805, 807, 819, 823, 831, 841, 843, 855, 867, 871, 873, 883, 885, 895, 897, 903, 907, 925, 927, 931, 933, 937, 957, 961, 975, 979, 981, 987, 991, 993, 997, 1009, 1011, 1021, 1023, 1029, 1033, 1039, 1041, 1051, 1053, 1057, 1083, 1087, 1093, 1095, 1101, 1105, 1107, 1113, 1117, 1123, 1125, 1137, 1147, 1155, 1167, 1179, 1183, 1189, 1197, 1201, 1203, 1209, 1213, 1219, 1231, 1233, 1243, 1245, 1249, 1251, 1261, 1263, 1267, 1275, 1281, 1285, 1291, 1293, 1297, 1303, 1309, 1311, 1315, 1323, 1329, 1339, 1345, 1353, 1357, 1359, 1365, 1369, 1371, 1377, 1387, 1389, 1393, 1395, 1401, 1407, 1411, 1417, 1419, 1423, 1431, 1435, 1437, 1441, 1449, 1453, 1455, 1459, 1461, 1465, 1471, 1473, 1477, 1479, 1483, 1485, 1489, 1491, 1493, 1495, 1497, 1499]

5. Family Portrait

Program Name: Family.java

Input File: family.dat

Judges Input File

```
2
1BBBBBBBBBB2
1AAA5678GGG2
123488889012
18888**88882
12888**88882
12777**77712
1234***9012
1234***9012
12888..88812
18888..88882
1((((__)))2
1(((__.))2
1---.---2
1$$$$$$$$$2
133556677882
1444aaa55aa2
1eerrttyuu2
1BBBBBBBBBB2

$$$$$$$$$$$
111111111111
222222222222
333333333333
444444444444
555555555555
aaaaaaaaaaaaa
CCCCCCCCCCCCC
EEEEEEEEEEEEE
GGGGGGGGGGGGG
HHHHH.HHH.HH
JJJJJJJJJJJJJ
666666666666
444444444444
777777777777
888888888888
999999999999
&&&&&&&&&&&
```

Judges Output to Screen (cont. on next page)

5. Family Portrait (cont.)

Judges Output to Screen

11111111111111111111
Be43\$- ((82222282AB
Be43\$- ((88337883AB
Br45\$- ((88447884AB
Bra5\$._ (88**78885B
Bta6\$._. . *****86B
Bta6\$-_. . *****87B
By57\$._)88**78888B
By57\$._)88997889GB
Bua8\$-))88007880GB
Bua8\$-))81111881GB
22222222222222222222

&98746JHGECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$
&98746J . GECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$
&98746J . GECa54321\$
&98746JHGECa54321\$
&98746JHGECa54321\$

6. Hash Codes

Program Name: HashCodes.java

Input File: hashcodes.dat

Judges Input File

```
7
3 12
4 22
3 65
3 10
5 13
4 23
4 35
```

Judges Output to Screen

```
7
34
14
4
0
39
181
```

7. Interlocked

Program Name: Interlocked.java

Input File: interlocked.dat

Judges Input File

```
11
bit one bointe
bit one bioonaet
bit one bierontt
bit byte bieyonte
computer program computerprogram
dog cat tacgod
dog cat gocoaadttoogg
log log log
golf tee gotleeamfet
swim suit simswuit
golf tee gotleeresmw
```

Judges Output to Screen

```
YES
YES
NO
YES
YES
NO
YES
YES
YES
NO
NO
```

8. Median Mile

Program Name: MedMile.java

Input File: medmile.dat

Judges Input File

8

MARY 21:12.32 22:14.45 23:55.84 24:48.99
ANN 28:16.39 28:10.87 27:04.00 27:05.01 29:29.71
LUCY 33:43.82 28:24.95 30:52.94 29:55.26
SALLY 24:12.42 26:14.85 28:59.04 27:04.67 28:52.99 26:59.87
SAMANTHA 20:23.45 23:34.45 21:34.98 22:23.90 23:12.34
RHONDA 20:34.89 21:59.99 25:34.98 23:11.78 22:59.00 22:49.00
LEIGH 28:18.39 25:10.87 26:04.00 27:25.21 27:15.71 26:52.24 26:52.76
RUTH 21:24.35 22:37.42

Judges Output to Screen

MARY 23:05.14
ANN 28:10.87
LUCY 30:24.10
SALLY 27:02.27
SAMANTHA 22:23.90
RHONDA 22:54.00
LEIGH 26:52.76
RUTH 22:00.88

9. Nautical Miles

Program Name: Nautical.java

Input File: nautical.dat

Judges Input File

8
1024
926
877
49999
1
9800
15000
72

Judges Output to Screen

1177.6
1064.9
1008.6
57498.9
1.2
11270.0
17250.0
82.8

10. Runners

Program Name: Runners.java

Input File: runners.dat

Judges Input File

```
5
5 12
*****
*.....R*T.**
**P**.....S**
*.....**
*****J*****
5 10
*****
**S*T...**
*.....J
*..P*R..**
*****
8 8
*****
*.R*****
*.....*
*....S.*
*..**.*
*...P..*
J..**.*T*
*****
6 10
*****J*****
*.T.....*
*S..**.*
*.....*
*R..**P..*
*****
8 10
*****
*T..*****
**.....*
**....P..*
*S.*..**.*
**.*....R*
**.....*
****J*****
```

Judges Output to Screen

```
10 RUSSELL
8 SETH THOMAS
9 RUSSELL
10 RUSSELL SETH THOMAS
10 RUSSELL SETH
```

11. Threes and Zeroes

Program Name: Threes.java

Input File: threes.dat

Judges Input File

```
10
4
5
12
38
7
14
15
17
49
6
```

Judges Output to Screen

```
300
30
300
330030
3003
30030
30
33303
3300003
30
```

12. Tree Decompression

Program Name: TreeDecompression.java

Input File: treedecompression.dat

Judges Input File

```
00000011010000001010001100100110001110011100
0000000001000000100100000010001000100111101000111
000000000010011010100111101001110010010101011001
000000110100000010010000101001000000001110
000000000010100110100001101001001010001010001101000101001000100100001
000001000001000110101010001000101010011010011001110100
```

Judges Output to Screen

```
ALFALFA
A DOG
MONKEY
AAAAH!
SCIENCE!!
#T#E#S#T#
```
